

A Super Wideband Miniaturized THz Antenna with a Slotted Partial Ground Plane for Ultra-high Speed 6G Communication Systems

Liton Chandra Paul

*Dept. of Electrical, Electronic and
Communication Engineering
Pabna University of Science and
Technology
Pabna 6600, Bangladesh
litonpaulete@pust.ac.bd*

Sayed Shifat Ahmed

*Dept. of Electrical, Electronic and
Communication Engineering
Pabna University of Science and
Technology
Pabna 6600, Bangladesh
shikor99@gmail.com*

Tithi Rani

*Dept. of Electronics and
Telecommunication Engineering
Rajshahi University of Engineering and
Technology
Rajshahi 6204, Bangladesh
tithirani@ieee.org*

Sk. A. Shezan

*Prince Faisal Centre for Renewable
Energy Studies and Applications, Dept.
of Electrical Engineering
Northern Border University
Arar, Saudi Arabia
shezan@nbu.edu.sa*

Md. Ashraful Haque

*Dept. of Electrical and Electronic
Engineering
Daffodil International University
Dhaka, Bangladesh
mdashraful.cce@diu.edu.bd*

Ali H Alenezi

*Dept. of Electrical Engineering
Northern Border University
Arar, Saudi Arabia
ali.hamdan@nbu.edu.sa*

Abstract— In this paper, we propose a novel design for a compact and efficient THz antenna, envisioned for prospective 6G communication. The miniaturized THz antenna architecture incorporates a new-shaped patch with rectangular slots, an inverted-L slots, and a slotted partial ground plane. The antenna is designed on a Rogers RT5880 material with a thickness of 10 μm and dimensions of $300 \times 300 \mu\text{m}^2$. The design achieves a super wide bandwidth of 620 GHz (2.83–3.45 THz) coupled with a low reflection coefficient and high radiation efficiency, peaking at 87%. Additionally, the antenna demonstrates commendable performance in terms of VSWR, directivity, gain, and radiation patterns. The analysis of radiation material variation (gold, graphene, and copper) reveals that copper offers the best results in terms of performance and cost effectiveness. The THz antenna is well-suited for a variety of 6G applications, including high-speed wireless communication, as well as imaging, sensing, security, and material characterization within the THz frequency spectrum.

Index Terms— Super wideband antenna; THz; Ultra-high speed; 6G communication.

I. INTRODUCTION

In recent years, global wireless data traffic has increased exponentially, driven by applications requiring data transmission rates of several tens of gigabits per second. According to Edholm's law, cellular data rates double approximately every 18 months [1]. Following the commercial debut of 5G wireless communication systems, research and industry stakeholders have identified 6G mobile communication systems as a promising and critical area of development. It is expected that 6G communication systems will be deployed around 2030, offering 10 to 100 times higher data rates, lower latency, wider coverage, and enhanced energy and spectrum efficiency compared to its 5G predecessor [2]. To achieve terabit-per-second (Tbps) capacities and seamless 6G connectivity, vast bandwidths are required. With limited spectrum, researchers explore THz and sub-THz (0.1–10 THz) frequencies, bridging microwave and optical bands with wavelengths of 0.03–3 mm. Despite their potential, the lack of efficient sources, detectors, and

interconnects has historically limited their widespread utilization [3], [4]. Compared to microwave frequencies, THz antennas offer several advantages, including ultra-wide bandwidth (supporting data rates up to 10 Gbps), improved spatial resolution, high directivity, low transmission power, and reduced interference. Additionally, they provide compactness, high efficiency, and strong directivity, making them well-suited for 6G communication [5], [6]. The THz spectrum has also gained significant attention due to its applications in ultra-fast wireless communication, remote sensing, spectroscopic detection, imaging, surveillance, material characterization and medical diagnostics [7].

As THz technology advances, short-range THz communication (up to 10m) is expected for indoor and outdoor use. However, the lack of compact, efficient THz sources remains a challenge, as current options do not meet size and power requirements [8]. Successful THz communication requires high-performance transmitters, receivers, and wideband antennas. Microstrip antennas, being compact, lightweight, and cost-effective, offer wide bandwidth, high efficiency, and reconfigurability [9]. Atmospheric path loss further hinders THz communication, necessitating efficient, low-loss antennas for stable performance [10].

To address the challenges associated with THz wireless communication, researchers have proposed various antenna designs to enhance efficiency, bandwidth, and applicability. The presented antenna in [7] offers a promising small-sized wideband (0.268 THz) THz antenna, which is suitable for indoor applications across multiple fields. However, the gold patch used with an area of $16,900 \mu\text{m}^2$, contributes to high fabrication costs. Many researchers have explored dual- and multi-frequency antennas for wireless communication systems [12], [14]. For instance, in [13], a dual-band THz (0.19–0.24 THz; 0.27–0.61 THz) Z-shaped monopole antenna was designed, offering promising features for short-range communication. In an effort to enhance THz communication

capabilities, [14] describes the modeling of a $1000 \times 1000 \times 200 \mu\text{m}^3$ antenna with a radiation efficiency of 55.88% operating in the 0.7–0.85 THz range, enabling more densely packed communication links and increased transmission security. Another approach involves the use of graphene-based THz antennas on a $20 \mu\text{m}$ dense polyimide, as demonstrated in [15], but with a narrow bandwidth (BW) of only 50 GHz. Additionally, researchers in [16] and [17] have investigated antennas for THz applications, employing Photonic Band Gap (PBG) structures to enhance performance. The simulation results in [18] indicate that a THz patch antenna with a $200 \mu\text{m}$ thick electromagnetic crystal substrate exhibits an ultra-wide bandwidth (350 GHz) and multi-frequency band operation. Moreover, for advanced THz communication, a single-pin-short antenna, featuring a multilayered substrate with dimensions of $400 \times 400 \times 455 \mu\text{m}^3$, was proposed in [19]. However, despite its potential, this design offers a relatively narrow bandwidth of 120 GHz. These research efforts highlight the continuous development of THz antennas to satisfy the growing demands of future 6G communication systems. However, further innovation is required to address the challenges of compact design, efficient radiation, and wideband operation in higher THz frequencies.

A super-wideband antenna, operating in the 2.83–3.45 THz range, is proposed for high-speed 6G communication, featuring a Rogers RT5880 substrate with a copper (Cu) patch and ground plane. The performance of copper, graphene, and gold as patch materials was compared. Results show that the copper patch achieves a return loss of -24.3 dB at 3.17 THz, a bandwidth of 620 GHz, a peak gain of 4.07 dB, and an efficiency of 87%. Performance values for graphene and gold were similar, with no significant deviations. A cost comparison indicates that copper is the most cost-effective, making it the optimal choice for real-time implementation. This ensures support for future 6G communication in the terahertz band. The paper is organized as follows: Section II discusses the geometrical structure of the proposed miniaturized THz antenna, Section III presents the simulation results along with a comparison table with previous works, and Section IV concludes the study.

II. ANTENNA DESIGN AND EVOLUTION

The geometrical structure of the proposed miniaturized THz wideband terahertz microstrip patch antenna, labeled with design parameters, is shown in Fig. 1. In the antenna, the substrate has a volume (length, width, and thickness) of $300 \times 300 \times 10 \mu\text{m}^3$. The copper material ($\sigma = 5.96 \times 10^7 \text{ S/m}$) with a thickness of $2 \mu\text{m}$ is used for designing the patch (top layer), which is mounted on the substrate and ground plane (bottom Cu layer) of the THz antenna. The substrate material is Rogers RT5880, with a dielectric constant of 2.2 and a loss tangent of 0.0009. Fig. 1(a) and (b) depict the front and rear views of the design, respectively. The structure consists of multiple arms, a square slot, and an inverted L slot in the radiating patch. The major geometrical parameters of the THz antenna are listed in Table I. The power feeding to the miniaturized THz antenna is achieved using a microstrip line

with a width of $30 \mu\text{m}$. This microstrip line is connected to a 50Ω SMA connector, ensuring optimal transmission of input power to the antenna and enabling efficient radiation. The microstrip feed line has dimensions of $80 \times 30 \times 2 \mu\text{m}^3$. All radiating parts of the antenna are designed using a single-layer copper. Fig. 2 provides a clear representation of the entire design process of the THz antenna, from the initial stage to the final stage. This diagram helps in understanding the steps involved in achieving the desired results. In the antenna, the microstrip line feeding is used, and the simulation and analysis has been accomplished using CST software.

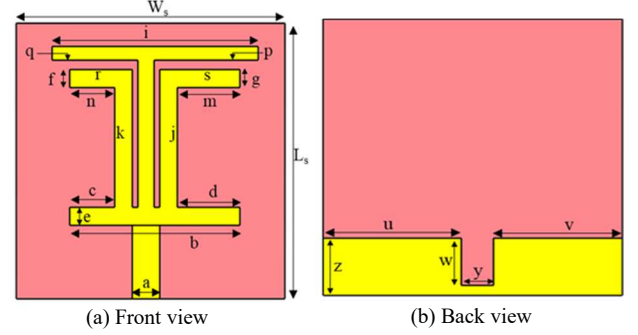


Fig. 1. Proposed miniaturized THz antenna

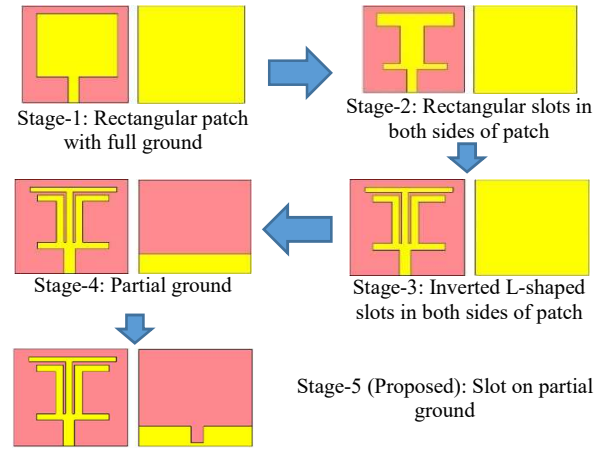


Fig. 2. Design flow of the proposed THz antenna

TABLE I. PHYSICAL DIMENSION OF THE THz ANTENNA

Symbol	Dimension (μm)	Symbol	Dimension (μm)
W_s	300	m	70
L_s	300	n	50
a	30	p	116
b	90	q	96
c	50	r	70
d	70	s	90
e	20	u	138.5
f	20	v	128.5
g	20	w	51
i	230	y	33
j	130	z	62
k	130	-	-

III. SIMULATED RESULTS AND ANALYSIS

The proposed antenna supports super wideband THz applications. Fig. 3 shows the reflection coefficient (dB) versus frequency curve for all five stages of the designed THz antenna. The Stage-5 (proposed) design demonstrates the best performance, achieving a significantly wider bandwidth compared to all other stages. For good impedance matching, the reflection coefficient of an antenna should be less than -10 dB. The super wide band THz antenna resonates at 3.17 THz with a reflection coefficient of -24.3 dB. It operates over a bandwidth of 620 GHz, spanning from 2.83 THz to 3.45 THz, with $S_{11} \leq -10$ dB.

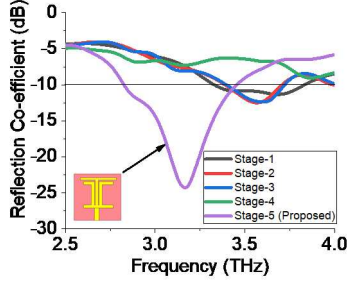


Fig. 3. Reflection co-efficient for all the stages of the THz antenna.

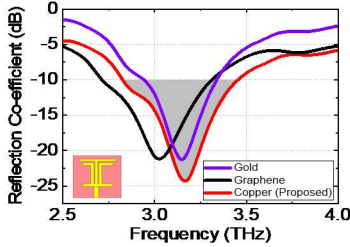


Fig. 4. Reflection co-efficient for different patch material.

The impact of gold, graphene, and copper as radiating materials for the designed antenna has been investigated as shown in Fig. 4, and copper demonstrates the best performance in terms of radiation characteristics and cost-effectiveness. In the frequency range of 2.83–3.45 THz, the imaginary part of the impedance is approximately zero, while the real part remains around 52.74Ω over a wider range, indicating a satisfactory impedance match. Due to impedance mismatches between the source and the antenna, the Voltage Standing Wave Ratio (VSWR) measures signal reflection from the antenna. For a well-structured antenna with proper impedance matching, the VSWR must be below 2, signifying reduced power reflection and minimal mismatch loss. The VSWR of the antenna remains below 2 within the super wide operating bandwidth of 2.83–3.45 THz, with a minimum VSWR of 1.1 at 3.17 THz. The polar plot in Fig. 5 illustrates the radiation pattern of the THz antenna, where the lowest side lobe levels for the E-field and H-field are -2.4 dB and -3.4 dB at $\phi = 0^\circ$ and $\phi = 90^\circ$, respectively. The maximum half-power beamwidth (HPBW) for the E-field and H-field is 81.3° at $\phi = 0^\circ$. Similarly, at $\phi = 90^\circ$, the HPBW is 19.8° . The magnitude of the main lobe at $\phi = 0^\circ$ for the electric and magnetic fields is 18 dBV/m and -33.5 dBA/m, respectively, directed at an angle of 175° . On the other hand, at $\phi = 90^\circ$, the main lobe is directed at an angle of 151° with a magnitude of 18.9 dBV/m for the E-field and -32.6 dBA/m for the H-field. The gain, directivity, and efficiency of the miniaturized THz antenna are shown in Fig. 6. At the chosen operating frequency of 3.17 THz, a respectable gain of 4.07 dB is observed. The directivity characteristics over the operating band reach a peak value of 6.06

dB at 3.17 THz. The radiation pattern of an antenna illustrates how its field strength varies with direction. The 3D radiation pattern of the THz antenna at the resonant frequency of 3.17 THz demonstrates that the proposed antenna is highly efficient and suitable for various applications in the terahertz region. The proposed THz antenna, which features a slotted ground plane, has its radiation efficiency shown in Fig. 6. Efficiency is defined as the ratio of gain to directivity. The maximum radiation efficiency is approximately 87%, with variations ranging from 81% to 87% across the entire band. This indicates that a substantial proportion of the input power, approximately 83.5% on average, is effectively radiated by the proposed THz antenna. This high level of radiation efficiency ensures optimal performance in terms of power radiation, making the antenna a promising candidate for 6G applications.

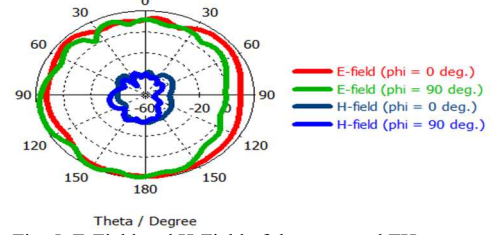


Fig. 5. E-Field and H-Field of the proposed THz antenna.

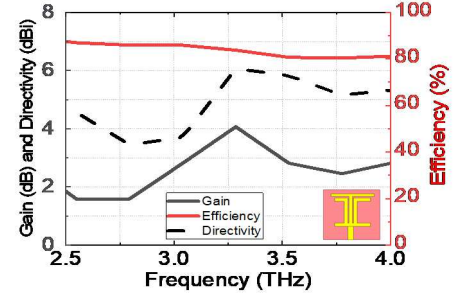


Fig. 6. Gain, directivity and efficiency of the THz antenna.

Fig. 7 shows the simulated distribution of surface current for the THz antenna at its resonating frequency of 3.17 THz. It can be observed that the surface currents are predominantly intensified along the lower patch of the feed line. This concentration is also noticeable at the peripheries of the arms of the miniaturized THz patch structure. This current distribution pattern plays a significant role in the resonant behavior of the THz antenna at the specified frequency.

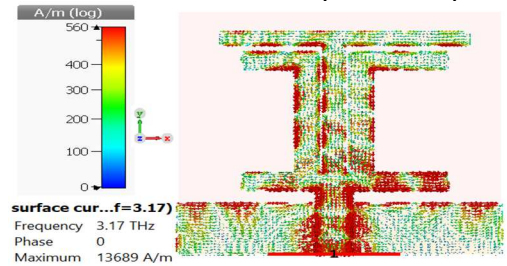


Fig. 7. Current distribution of the THz antenna

The proposed THz antenna demonstrates significant miniaturization and a wide operating BW of 620 GHz, along with notable efficiency, when compared to previously reported antennas, as shown in Table II. This performance advantage can be attributed to the carefully optimized dimensions and the use of low-cost copper, which contributes

to the antenna's gain and directivity. A comprehensive comparison has been made with other antennas cited in the literature, highlighting the superior characteristics of the proposed design. The antenna's enhanced performance has garnered considerable attention, positioning it as a promising candidate for future 6G systems, which are expected to be deployed by 2030.

TABLE II. COMPARISON TABLE

Ref No.	Size (L×W×h) μm^3	Sub. Material	Patch Material	BW (GHz)	Max. Effi. (%)	Applications
7	300×300×45	Polyimide	Gold	269	97.3	High-Speed Short-Range Wireless Communication
11	460×400	SiO ₂	Graphene	60	35	THz-band Represent
12	500×500×10	Pyrex	-	200	-	Wireless Communications
13	850×770×197	Rogers RT6006	Copper	50, 340	-	THz Communication
14	1000×1000×200	RT/Duriod 6006	-	150	55.88	Short Distance Wireless Communication
15	208.98×433.2×20	Polyimide	Graphene	50	86.58	THz-band
16	800×600×191.29	Polyimide	Copper	34.92	82.46	Terahertz Spectral Band
		PBG		36.23	85.71	
17	600×600×90	Polyimide	Graphene	300	90.84	Sensing and Communication Technologies
18	1000×1000×200	Electromagnetic Crystal	-	350	67.02, 76.45, 88.30	Modern Communication
19	400×400×455	-	-	120	86.36, 84.27	Advanced Wireless Communication
Proposed	300×300×10	Rogers RT5880	Copper	620	87	Ultra-high Speed 6G Communication

IV. CONCLUSION AND DISCUSSION

This paper presents a proposal for a highly efficient, super wideband, compact THz antenna designed to support promising applications in the THz regime, particularly for future 6G communications. The conventional antenna structure has been modified by incorporating rectangular slots, inverted-L slots on the patch surface, and a slotted partial ground plane. These modifications have resulted in wideband resonance characteristics, specifically achieving a bandwidth of 620 GHz (2.83–3.45 THz) while maintaining $S_{11} \leq -10$ dB. The antenna structure is developed using a 10 μm thick Rogers RT5880 substrate (300×300 μm^2), with copper as the conducting material for both the patch and ground layer surfaces. The operating band enables the antenna to achieve a maximum peak radiation efficiency of 87%. Additionally, the proposed THz antenna exhibits notable characteristics, including a VSWR of 1.1, good directivity, a gain value of 4.07 dB, and efficient radiation patterns. Based on these achievements, it can be concluded that the proposed wideband THz antenna holds significant potential for various 6G applications. These include, but are not limited to, high-speed wireless communication, biomedical imaging systems,

sensing capabilities, security scanning, explosive detection, and material characterization within the terahertz frequency band.

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